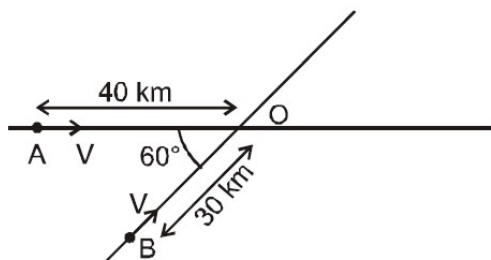
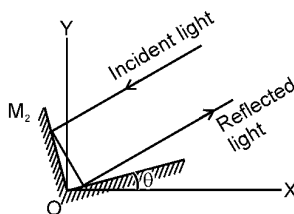


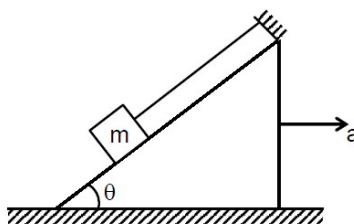
1. Two cars A and B moving on two straight tracks inclined at an angle 60° heading towards the crossing initially their positions are as shown in the figure. Both cars have same speed. Minimum separation between them during their motion will be.



- (A) 10 km (B) $5\sqrt{3}$ km (C) 5 km (D) $\frac{20}{\sqrt{3}}$ km
2. A light ray gets reflected from a pair of mutually $\perp$ mirrors, not necessarily along axes. The intersection point of mirrors is at origin. The incident light is along $y = x + 2$. If the light ray strikes both mirrors in succession, then it may get reflected finally along the line :

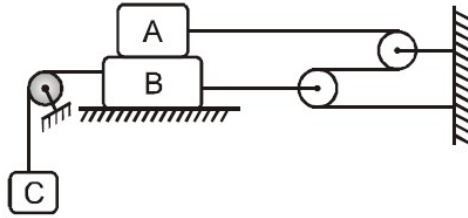


- (A) $y = 2x - 2$ (B) $y = -x + 2$ (C) $y = -x - 2$ (D) $y = x - 4$
3. The wedge is moved towards right with acceleration a , which is slowly increased from zero. The tension in the string just when the small block of mass m breaks contact with the surface of wedge is (Assume all surfaces smooth, the string inextensible and neglect rotational effect's.)

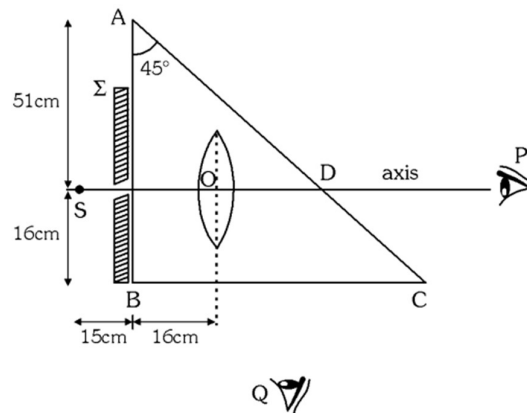


- (A) $mg \sin \theta$ (B) $mg \operatorname{cosec} \theta$ (C) $mg \tan \theta$ (D) $mg \cot \theta$

4. Block A of weight 500 N and block B of weight 700 N are connected by rope pulley system as shown. The largest weight C that can be suspended without moving block A and B is W. The coefficient of friction for all plane surfaces of contact is 0.3. The pulleys are ideal. Find $\frac{W}{90}$.

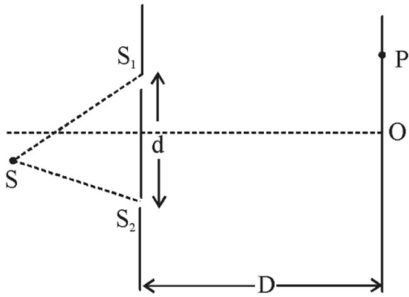


- (A) 12 (B) 22 (C) 9 (D) 18
5. An isosceles right angled triangular glass ($\mu = 1.6$) prism has a cavity inside it in the shape of a thin convex lens whose both surfaces have radius of curvature equal to 20 cm. The cavity has been filled with a transparent liquid of refractive index 2.4. S is a point source and Σ is an opaque sheet having a small hole such that the source and the hole both lie on the principal axis of the lens. The small hole in the opaque sheet is just to ensure that only paraxial rays are incident on the optical system. However, the size of hole is large enough to neglect diffraction effects.

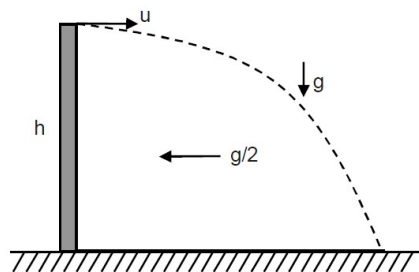


- Choose the correct statement:
- (A) The observer P will see the image at a distance of 2cm from D on line SOD.
 (B) The observer P will see the image at a distance of 1cm from D on line SOD.
 (C) The observer Q will see the image at perpendicular distance of 6.9cm from line BC.
 (D) The observer Q will see the image at perpendicular distance of 10.9cm from line BC.
6. A block of mass m is pulled on an incline surface having coefficient of friction $\mu = 1$ & angle of inclination $\theta = 30^\circ$, with the horizontal, such that required external force is minimum. The angle made by this force with the incline is :
- (A) 45° (B) 30° (C) 75° (D) 53°

7. A particle undergoes from position $O(0, 0, 0)$ to $A(a, 2a, 0)$ via path $y = \frac{2x^2}{a}$ in x - y plane under the action of a force which varies with particle's (x, y, z) coordinate as $\vec{F} = x^2y\hat{i} + yz^2e^{2z}\hat{j} - \left(\frac{z}{x+2y}\right)\hat{k}$. Work done by the force $\vec{F}$ is: (all symbols have their usual meaning and they are in SI unit.)
- (A) $\frac{4a^4}{5}$ (B) $\frac{a^4}{5}$ (C) $\frac{a^3}{4}$ (D) $\frac{2a^4}{5}$
8. It is given that $SS_1 - SS_2 = \frac{\lambda}{2}$ and $S_2P - S_1P = \frac{\lambda}{2}$ Intensity from slit S_2 is I_0 then which are CORRECT option.



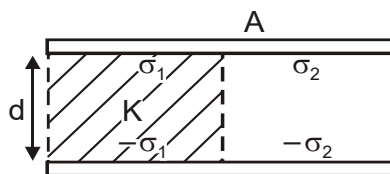
- (A) Central maxima will form at point P (B) Central maxima will form at point O.
 (C) Resultant intensity at P is $4I_0$. (D) Resultant intensity at P is I_0 .
9. The position of a particle moving along x -axis depends on time as $x = 2(1 - e^{-3t})$. Choose the correct statement(s).
- (A) Total displacement of the particle is 2 units.
 (B) Total distance covered by particle is 2 units.
 (C) Velocity of the particle will change its direction at some instant.
 (D) Velocity of particle will not change its direction throughout the motion.
10. A particle of mass m is projected horizontally from a tower with velocity u towards East. Wind provides constant horizontal acceleration $\frac{g}{2}$ towards west. If particle strikes the ground vertically then the :



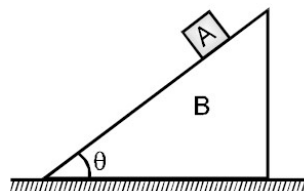
- (A) Height of tower is $\frac{2u^2}{g}$
 (B) The speed with which particle strikes ground is $2u$
 (C) Work done by wind force $\frac{1}{2}mu^2$
 (D) Particle strikes the ground after time $t = 2u/g$

11. A particle moving with constant acceleration on a straight line covers 20 m in 7th sec and 24 m in 9th sec. Choose the correct options regarding motion of particle.
- (A) Acceleration of the particle is 2 m/s²
 (B) Acceleration of the particle is 4 m/s²
 (C) Distance travelled by particle in 10th sec is 26 m.
 (D) Distance travelled by particle in 10th sec is 30 m.

12. A parallel plate capacitor of area A and separation d is charged to potential difference V and removed from the charging source. A dielectric slab of constant $K = 2$, thickness d and area $\frac{A}{2}$ is inserted, as shown in the figure. Let σ_1 be free charge density at the conductor-dielectric surface and σ_2 be the charge density at the conductor-vacuum surface.

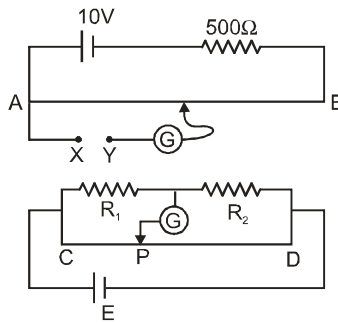


- (A) The electric field have the same value inside the dielectric as in the free space between the plates.
 (B) The ratio $\frac{\sigma_1}{\sigma_2}$ is equal to $\frac{2}{1}$.
 (C) The new capacitance is $\frac{3\epsilon_0 A}{2d}$
 (D) The new potential difference is $\frac{2}{3} V$
13. In the figure shown, A & B are free to move. All the surfaces are smooth. ($0 < \theta < 90^\circ$)

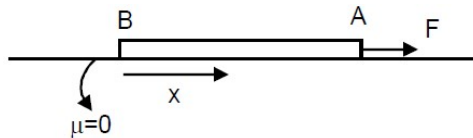


- (A) the acceleration of A will be more than $g \sin \theta$
 (B) the acceleration of A will be less than $g \sin \theta$
 (C) normal force on A due to B will be more than $mg \cos \theta$
 (D) normal force on A due to B will be less than $mg \cos \theta$

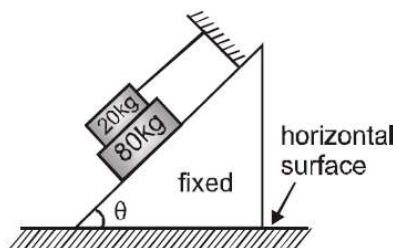
14. In the circuit shown AB is potentiometer wire of length 10m and resistance 500Ω . CD is a 1m wire of meter bridge which is balanced at CP = 20 cm. Given that when the potential difference across R_1 is applied across XY the balancing length on potentiometer is 2m and for potential difference across R_2 , the corresponding length is 8m. Then



- (A) $\frac{R_1}{R_2} = \frac{1}{4}$ (B) Potential difference across R_1 is 1V
(C) Potential difference across R_2 is 4V (D) EMF of the battery $E = 5V$
15. A non-uniform rope of length L is placed on the ground and Rope is pulled by a constant force F as shown in the diagram. There is no friction anywhere, linear mass density of rope is given by $\lambda = \lambda_0 \left(1 + \frac{x}{L}\right)$, where x is measured from end B. Tension at a distance of x from B is T , total mass of the rope is M and acceleration of rope is a_0 Then Choose the CORRECT option(s)



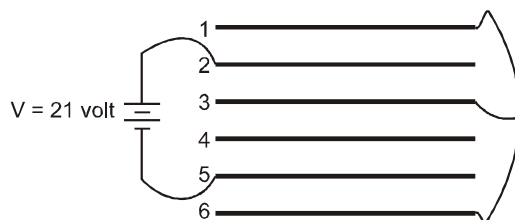
- (A) $\int_0^L T dx = \frac{4FL}{9}$ (B) $M = \frac{3\lambda_0 L}{2}$ (C) $a_0 = \frac{2F}{3\lambda_0 L}$ (D) $T = \frac{2F}{3L} \left[x + \frac{x^2}{2L} \right]$
16. The system shown is in limiting equilibrium. The coefficient of friction for all contact surfaces is $\frac{1}{4}$



- (A) $\tan \theta = \frac{3}{8}$
(B) Tension in the string $= \left(\frac{100}{3} g \sin \theta \right) N$
(C) Net frictional force on 80 kg block is $(80 g \sin \theta) N$
(D) Force exerted by 20 kg block on 80 kg block is $(20 g \cos \theta)$

Paragraph for Question Nos. 17 to 18

Six identical metal plates of area A each are arranged horizontally parallel to each other, with equal gaps d. Counted from the top, they are numbered 1, 2, 3,, 6. Plates 1, 3 and 6 are connected together. A potential difference of 21 volts is applied between plates 2 and 5. Take value of $\frac{\epsilon_0 A}{d}$ to be equal to $7\mu\text{F}$



17. Charge supplied by battery to the system is :
 (A) $252\ \mu\text{C}$ (B) $126\ \mu\text{C}$ (C) $84\ \mu\text{C}$ (D) $142\ \mu\text{C}$
18. Charge on lower face of plate 4 is :
 (A) $184\ \mu\text{C}$ (B) $84\ \mu\text{C}$ (C) $42\ \mu\text{C}$ (D) zero
19. Intensity observed in an interference pattern $I = I_0 \sin^2\theta$. At $\theta = 30^\circ$ intensity $I = 5 \pm 0.002$. The percentage error in angle if $I_0 = 20 \frac{\text{W}}{\text{m}^2}$ is $\frac{n\sqrt{3}}{\pi} \times 10^{-2}\%$. Find n.
20. In the young's double slit experiment a point P from the central bright fringe is such that the intensity at the point P is $\frac{3}{4}$ times the maximum intensity. Distance between the slits is d and wavelength is λ . The angular position of P is $\sin^{-1} \sin^{-1} \left(\frac{\lambda}{nd} \right)$. Find n.

ANSWER KEY

1.	(B)	2.	(D)	3.	(B)	4.	(C)	5.	(C)
6.	(A)	7.	(D)	8.	(A)	9.	(ABD)	10.	(ABD)
11.	(AC)	12.	(ABCD)	13.	(AD)	14.	(ABCD)		
15.	(ABCD)	16.	(ABC)	17.	(B)	18.	(C)	19.	4
20.	6								